

AP CSP | Lesson 05: Sampling & Representation

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Answer all 12 questions. Q5 requires exactly TWO selections. Show calculations with units. Pixel and sample payload questions exclude metadata, compression and padding unless explicitly stated.

1. A sensor sends 00101100 as an unsigned binary integer. Which decimal value does the receiving program obtain? [1 mark]

A. 40 B. 42 C. 44 D. 48

Answer: _____

2. A program executes: $\text{samples} \leftarrow 4$; $\text{samples} \leftarrow \text{samples} * 2$; $\text{bits} \leftarrow \text{samples} * 3$. What value is stored in bits? [1 mark]

A. 12 B. 14 C. 24 D. 32

Answer: _____

3. A sensor changes from one measurement every 2 seconds to one every 0.5 seconds. How does its sampling rate change? [1 mark]

A. It becomes one quarter as large. B. It doubles.
C. It stays the same. D. It becomes four times as large.

Answer: _____

4. A toy sensor stores only the levels 0, 3, 6 and 9. It rounds to the nearest level. Which value is stored for a measurement of 5.2? [1 mark]

A. 3 B. 5.2 C. 6 D. 9

Answer: _____

5. A recorder stores fixed-length sample codes. Which TWO changes double its uncompressed sample payload? Consider each change separately; all other settings stay fixed. Select exactly TWO answers. [1 mark]

A. Double the number of samples. B. Double the number of possible value codes from 8 to 16.
C. Double the bits stored per sample. D. Halve the recording duration at the same sample rate.

Answer: _____

6. A bitmap contains 10×6 pixels using 4 bits per pixel. With no metadata, padding or compression, how many bytes of pixel data are stored? [1 mark]

A. 24 B. 30 C. 60 D. 240

Answer: _____

AP CSP | Lesson 05: Sampling & Representation

Written responses | Questions 7-9

7. [4 marks]

A smooth curve models temperature through a day. A file records a temperature rounded to the nearest whole degree every hour.

- (a) Classify each representation as analog or digital.
- (b) Explain why the file is an abstraction.
- (c) Identify two kinds of detail the file can omit.

8. [4 marks]

A toy recorder measures 1.1, 3.8, 6.2 and 8.4 at equally spaced times. It stores the nearest level from {0, 3, 6, 9}.

- (a) Give the four stored values in order.
- (b) State the minimum bits per sample.
- (c) Find the payload for these four samples, with no overhead.

9. [4 marks]

A visual sampling model offers 4, 8 or 16 samples over the same time window, and 4, 8 or 16 value levels over the same range.

- (a) Which setting directly helps reveal changes between measurement times?
- (b) Which directly helps distinguish nearby measured values?
- (c) At 16 samples and 8 levels, find minimum bits per sample and total payload.

AP CSP | Lesson 05: Sampling & Representation

Written responses | Questions 10-12

10. [4 marks]

A custom code assigns 00=A, 01=B, 10=C and 11=space.

- (a) Decode 01 00 11 10.
- (b) Encode "CA" with no space.
- (c) Can a receiver reliably interpret this stream without knowing its coding rules? Explain.
- (d) Does this example establish that every real text character uses two bits?

11. [4 marks]

A bitmap is 20 pixels wide and 12 pixels high, using 2 bits per pixel. Ignore metadata, padding and compression.

- (a) Find the number of pixels.
- (b) Find the pixel payload in bits and bytes.
- (c) Both dimensions double while bits per pixel stay fixed. By what factor does payload grow?

12. [4 marks]

A student says, "Changing a recording from 8 to 16 samples makes it an exact copy and costs no more storage." Bits per sample and recording duration stay fixed.

- (a) Explain one possible benefit.
- (b) Explain why exactness is not guaranteed.
- (c) Compare the uncompressed sample payloads.
- (d) Name a different setting that can improve value precision over a fixed range.
